I have been given explicit permission by the Director of Network and Security/CISO of Connecticut College, John Schaeffer, to use this as a work sample.

Network Operations Assistant Guide

For

Connecticut College

Created by

Evangeline Wong

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Guide Purpose

This guide serves as a handbook and means of knowledge transfer for Network Operations Assistants working at Connecticut College for the summer, written by Evangeline Wong, a past Network Operations Assistant.

This guide will improve team efficiency by providing convenient and easily accessible resources for Network Operations Assistants beginning their internship and has resources as they progress throughout their internship.

While this guide is not an exhaustive list to prepare interns for every eventuality of their time as a Network Operations Assistant it will provide a foundation for those starting out and insights and lessons learned from past interns.

It is encouraged for present interns to create their own iteration of this guide for the next intern to learn from so that this guide can be improved upon and knowledge from each past intern can be passed down, while also giving credit to past interns who have made previous editions of this guide.

Should an intern choose to create a guide of their own it is ***highly recommended*** to ***only*** work on this guide during working hours and ***not*** at home. This is so that interns will be paid for their time working on this guide.

If you have any questions regarding your employment at the college please refer to the Connecticut College employee handbook linked below or reach out to HR at humanresources@conncoll.edu or walk to Strickland house and see if anyone is free to answer your questions or set up an appointment with someone there.

https://drive.google.com/file/d/1x0Kv7vRpICr2Bz0gvVsk_Kh1pdbnyUiH/view

Day 1 Suggested Tasks

1. **Acquire Camel Card ID**
    - **Context**
        - It is important to have a Camel Card ID on you and to start the process of getting your card on the first day of internship because the card is what is going to easily allow you access to the dorm buildings.
        - It's easier to swipe your Camel ID versus finding specific doors with a keyhole to get in with your NIS keys or having to call campus safety and have them buzz you in.

        - And some dorm buildings don’t have any doors that have keyholes and can only be accessed with a Camel ID. So it is imperative to get your Camel Card ID quickly.
    - **Steps to Obtain:**
        - Upload a photo of yourself to Camel Web:
            - Go to Camel Web > HR/Finance > Camel Card > Menu > Submit New Photo.
        - Ensure your photo meets the requirements for an approved ID photo.
        - Wait a few days for campus security to approve it.
        - Check back on Camel Card to view progress on photo approval.
        - After approval, go to Card Services during their open hours (check Camel Web for open hours) and pick up your Camel Card ID.
        - In the days waiting for your Camel Card, if you need access into a dorm building either use your NIS keys and find a door with a keyhole and use Key A or wait by a door with a card tap device and call Campus Safety to remotely open the door for you by letting them know what part of the building you are at and your name and job title.
        - Campus Safety number: 860-439-2222
2. **Check NIS Keys Access**
    - **Context**
        - Equally important to having a Camel ID is a set of NIS keys. The importance of these keys is that they allow entry into all dorm rooms to do your job of inspecting and repairing broken network jacks in dorm rooms.
        - And they allow access to your office when you arrive in the morning.
        - And they are a back up to get into dorm buildings if there is a side door with a keyhole.
        - And they are needed to enter Data Rooms where the switches are.
        - A Camel ID cannot grant you access to a Data Room only a NIS Key A will.
    - Make sure you are on the list to check out NIS keys from the guard shack.
    - Speak to the Campus Safety Officer at the Guard Shack about this.
    - Inform them that you are a Network Operations Assistant.
    - They might call Kevin or John to confirm.
    - NIS Key A grants access to Data Rooms where the switches are and to other staff/facility and office rooms.
    - NIS Key B grants access to dorm rooms.
3. **Begin Inspections**
    - **Context**

        - The very first day will most likely be getting acquainted with your internship and the people you will be working with. You will be shown by someone else on the team how to do a dorm inspection and you will shadow them. Additionally you may be shown other things as well.
    - By day 2 or 3, depending on what is going on in your internship, you should be able to go out on your own and begin dorm inspections. Confirm at the beginning of the day if you can go out on your own to do dorm inspections or if there is something else planned for you.
    - Leave a message in the NSS chat about which dorms you will be going to before you start. This way, your coworkers will know where you will be going for safety reasons.
4. **Download Google Chat to Your Phone**
    - Google Chat will be the main mode of communication between you and your coworkers and managers while out in the field.

Points of Note

- **Start and End Times:**
    - Start: 830am
    - End: Summer Hours 4pm or if working into academic year 5pm
    - Lunch: 12:00 pm 30 mins unpaid. Input manually in Dayforce Timesheet. You will also be allowed to eat at the dining halls for a price. Please confirm with others the cost.
- **Meetings:**
    - Usually in-person check-in meetings are only on Mondays. Every other meeting is virtual via Google. Sometimes there are no meetings. Check the NSS group chat to see if there is a meeting. They will only report if there is going to be a meeting. Sometimes in person meetings happen on other days besides mondays.
    - If there is a meeting, Google Chat will notify you about it.
- **Tech Check-Ins:**
    - Every other Friday via Google Meet unless there is not a meeting.
    - Relay what you have done and are currently working on.
- **Transportation:**
    - If you have a car, use it to transport heavy items between buildings.
    - A golf cart may be available for you to drive around campus if it is available. Check with other coworkers for availability.
- **Hydration**
    - Recommended that you stay well hydrated before, during, and after your shift because walking around campus and in and out of dorm buildings and rooms during summer is sweaty work.

Holidays, Special Events on Campus, and Intern Perks

Overview

As a summer intern, it's important to be aware of holidays, special events on campus, and intern perks that may affect your work schedule or provide opportunities to enjoy campus life. This section will guide you on how to navigate these occasions and make the most of the perks that come with them.

Holidays to Be Aware Of

1. 4th of July:
    - The 4th of July is a federal holiday, and you are not required to work on this day.
    - Depending on what day the 4th is on you may get an extra day off from work.
    - It is recommended to ask a coworker or manager to confirm what days you will have off.
    - For example, during my internship, the 4th, was on a Thursday and I also had the following Friday the 5th off. Something similar may happen in your internship. It is best to confirm with others around if something similar is in your situation.
2. Labor Day:
    - Labor Day is another federal holiday when the campus will be closed, and you won’t be expected to work. Make sure to plan accordingly, especially if you have ongoing projects that need attention before the long weekend.
3. Checking Your Work Schedule:
    - As an intern, it’s important to confirm whether you are expected to work on federal holidays or if the campus will be closed. If you’re unsure, check with your supervisor ahead of time. Typically, you will not be required to work on these holidays, but it’s always best to verify.

Campus Events and Free Food

1. Stay Informed via Conn Email:
    - Regularly check your Connecticut College email account. The college often sends out notifications about upcoming campus events, some of which may offer free food. These events can be a great way to enjoy a meal, meet new people, and experience campus life.
2. Special Events:
    - Throughout the summer, the college may host various special events, such as guest lectures or celebrations. These events sometimes include free food and are open to staff, including interns.

    - Make a habit of scanning your emails for these opportunities. Participating in these events can enrich your experience and provide some nice perks, like a free lunch or snacks.

Tips for Navigating Holidays and Events

- Plan Your Work Around Holidays: If you know a holiday is coming up, make sure you complete any urgent tasks beforehand. This will help you avoid last-minute stress and ensure that you can enjoy the holiday without worrying about unfinished work.
- Take Advantage of Free Campus Events: Free food events are a great perk of being on campus. They not only save you money but also offer a chance to engage with the college community and take a break from your daily routine.

Other Intern Perks

During your internship you will also be able to use the college's gym for free and check out books at Shain Library. Check hours of operation on CamelWeb. Please note that you will need a Camel Card ID for access to these perks.

Parking Permit

- If your internship goes into the academic year then you will need a parking permit on your car by September 1st.
- I recommend ordering yours online at least a week before September 1st
- Steps to obtain parking permit:
    - Login into Camel Web
    - Select My Conn
    - Select HR/Finances
    - Scroll down and select Parking Permits and Appeals
    - Select Affiliated Login
    - Select Get Permit
    - Follow the steps to obtain a parking permit
    - A parking permit for Faculty and Staff is free
    - Once you have completed the steps to ordering your parking permit a confirmation email will be sent to you and it will say it will be delivered to your mailbox in Bill Hall. In my experience it was not delivered to my mailbox and I went to Camel Card Services to pick my parking permit up. You may have to do the same.

Start of Day Tasks

1. **Check Out NIS Keys:**
    - Stop by the Guard Shack on your way in and ask to check out NIS keys.

   - Remember to return the keys before leaving at the end of the day.
2. **Login Timesheet Hours:**
   - Login to Camel Web: Camel Web
   - Navigate: My Conn > HR/Finance > Your Employment Details > Dayforce Timekeeping System
   - Select 'Employee Timesheet'
   - Log your start and end time and your lunch time, then click the green check marks.
   - Click 'Save' on the upper left-hand side of the screen.
3. **Check Emails, Web Help Desk, & Google Chat:**
   - I recommend waiting until at least 9:30 AM before leaving your office for dorm checks/repairs/other out of office tasks to ensure there isn't a morning meeting on Google Meet.
   - In the meantime, do assigned tasks from Web Help Desk or check for new tickets. If there's nothing to do, study your coursework, study network topics, or ask around if there is anything you can help with.

Downtime

When you have downtime I recommend reflecting on any lessons you have learned throughout your internship. Ask or research questions that you have. And keep a log of tasks you have done so you can add them to your resume for future internships or job applications. You can also study for classes you will be taking in the fall or study network concepts that are also included in this guide.

End of Day Tasks

1. Always make sure the tools that you are using, if borrowed from someone else, are kept in your office.
2. Don't take tools that aren't yours home.
3. Make sure your office is kept organized and you know where all your tools and repair supplies are.
4. Lock your office door on the way out.
5. Return your NIS keys to the guard shack.

Suggested Daily Tools to Bring at Beginning of the Day

- Small pocket sized notepad and pen to record notes of things learned.
- Headphones with microphone for virtual meetings. Webcam optional.
- Homework or study materials for downtime.
- Hat, sunglasses, water, umbrella, sunscreen, backpack.
- Lunch, snacks, comfortable clothing, and walking shoes.

- A laptop to do tickets you have been assigned. Ask coworkers if there is a spare student laptop that you can have on standby in case you need to bring one with you.

How to Login to Web Help Desk

- **Recommended Browser:** Microsoft Edge
- **Login Link:** Web Help Desk
- **Credentials:**
  - Username: stuadminacct
  - Password: ask coworkers or managers

Dorm Checks and Fixing Network Jacks

Overview

This section provides guidance on how to perform dorm checks and repair network jacks during your internship. You can approach this task in different ways depending on your level of experience and comfort with the work.

Approaches to Dorm Checks and Repairs

1. **Sequential Approach** (Ideal for Beginners)
   - **First 1-2 Weeks**: Focus solely on dorm checks without doing any repairs. This allows you to become familiar with the process and the layout of the dorms.
   - **Subsequent Weeks**: Start combining checks with repairs as you gain confidence.
2. **Combined Approach** (For Those with Experience)
   - Perform dorm checks and repairs simultaneously as you move through the dorms. This method is more efficient and helps ensure that repairs are completed in a timely manner.
3. **Separate Approach** (Optional)
   - Conduct all dorm checks first, then return later to complete any necessary repairs.

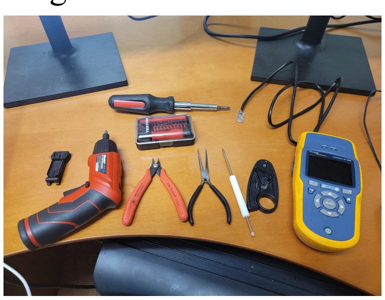
Suggested Method

- **Beginner Recommendation**: If you have no experience with fixing network jacks, spend at least the first week only doing dorm checks.
- **Experienced Technicians**: If you are already comfortable with repairs, fixing issues as you check dorms is the most efficient option.

How I Started

- Initially, I focused only on dorm checks.
- Later, I transitioned to checking and repairing simultaneously. I found this method to save time and ensure all tasks were completed before the fall semester began.

Equipment and Supplies

- **Carrying Equipment**: The tools needed for repairs are light enough to carry in your backpack as you move through the dorms. Ensure your backpack is stocked with enough supplies at the start of each day.
    - Start of Day Supplies:
        - 10 Faceplates
        - 20 Network Jack Holders
        - Bag of extra screws
        - Tools Pictured below
        - At least two 1ft patch cables
        - At least two 3ft patch cables
        - Fluke Network Device
        - Network Tone Device
- **Extra Supplies:** Extra supplies you will need to do network jack repairs will be located in the supply room 104 in Bill Hall which needs a Camel Card to swipe for access or the old office space in room 109 if that is still empty which needs NIS Key A to access.
- **Tool Selection**: If borrowing network repair tools, such as from Kevin Northcutt, take only the tools you need (pictured below) and set aside the rest to reduce unnecessary weight.
- 

Environmental Considerations

- **Weather Conditions**: Dorm checks can be hot and humid depending on the weather. Take regular breaks and drink plenty of water to stay hydrated.

- **Working in Occupied Dorms**: If dorm checks extend into the period when students have moved back in, you may be able to knock on doors and ask to inspect/repair network jacks, but confirm before proceeding.
- It may not be an efficient use of your time to knock on dorm room doors since you don't know the schedule of the student living in the dorms.
- It may be better to wait for a ticket to be assigned to you if a student from a dorm you have not done an inspection on yet.
- This way you can schedule a time that works for both of you to do the repair.
- As long as access to each dorm building is available before students move back in, you shouldn't have to worry about having to inspect network jacks in dorm rooms after students have moved in.

Special Cases and Where to Start Inspecting First

- **River Ridge Apartments**:I recommend starting at River Ridge first and inspecting and fixing those network jacks. And If you plan to repair a network jack on a different day than the inspection, make sure to note the exact room and location of the broken jack.

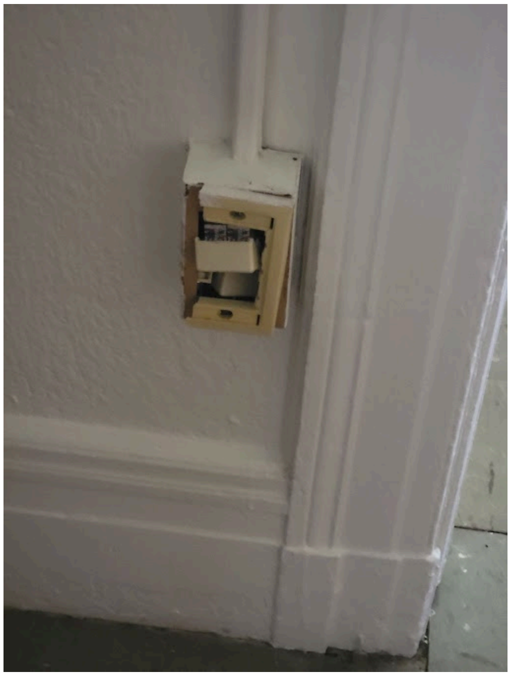
Additional Notes

- **Supply Management**: It's unlikely you'll run out of supplies before lunch if you start with enough at the beginning of the day.
- **Guidance on Repairs**: A detailed guide on how to repair network jacks is included in this document.
  - And while out in the field alone don't hesitate to ask questions or for guidance via the NSS chat you will be added into or individually message your coworkers for help or questions.

Types of Network Jacks that Need Fixing

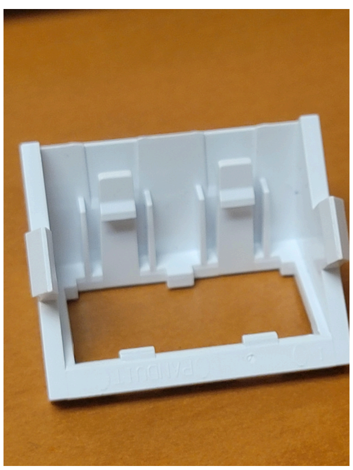
<u>No blue network port available.</u>

- The face plate needs to be removed and the inside checked if the blue network jack is inside.
- There usually always is a blue network cable on the inside of the box if a blue network jack is not immediately visible on the faceplate.
- When working on a network box, always test with the Fluke Network Device to make sure the jack is working properly and the cable is wired correctly.
- A Guide on how to use the Fluke Network Device is included in this guide in a different chapter



<u>The sloped modules are pushed in or broken</u>

- The faceplace and the sloped holders will need to be removed.
- Check the sloped holder to see if there is any damage to them. If there is damage they will not be able to hold network jacks.
- This is what a functional holder looks like:



Loose faceplate
Removing the screw coverings and screwing the face plates back into position is enough.

Missing faceplate
Replace the faceplate and sloped module holders.

<u>Cracked faceplate</u>
Replace the faceplate and sloped module holders.

Practical Tips for Working in Dorms and Campus Apartments

Pets in Campus Apartments

Some of the dorms you’ll work in are actually on-campus apartments where residents may have pets. While it might be tempting to interact with these animals, it’s best to avoid petting or engaging with them. You never know the pet's temperament, history, or whether they might have issues with strangers. It’s important to maintain a professional boundary to avoid any potential incidents.

Handling Trash in Dorms

When you’re repairing a network jack in an empty dorm room, be mindful of where you dispose of any trash. Facilities staff prefer that the dorm room trash bins remain empty for when students move in. Instead of using the room’s trash bin, throw away any trash in the bins located in the bathrooms or hallways. This small step helps keep the rooms ready for incoming students and maintains a clean environment.

Construction in Dorms During Summer
During the summer, some dorm rooms may have ongoing construction. If you encounter such a situation, it's best to avoid those rooms for safety reasons. If a construction worker questions why you're in a dorm building, they might mistake you for a student, not knowing you're an intern. In such cases, simply inform them that you are a network operations intern and that it's your job to be in the dorm building to repair network jacks. Clear communication can prevent misunderstandings and ensure that you can carry out your work without issues.

Interacting with Campus Facilities Workers
Campus facilities workers, such as janitors or cleaners, might also mistake you for a student, especially if you're working in a dorm room or building during the summer. They might question your presence or initially be hesitant about letting you stay. If this happens, calmly explain that you are a summer intern and that your job involves being in the dorm buildings to carry out network repairs. In my experience, this approach clears up any misunderstandings, as they likely assumed I was a student.

These practical tips will help you navigate some of the unique aspects of working in dorms and campus apartments, ensuring that you maintain a professional and considerate approach in your day-to-day tasks.

Practicing Cat6 Cable Wiring Using the T568B Standard

Overview
As a network operations intern, one of the essential skills you'll need is the ability to properly wire network cables, specifically Cat6 cables, according to industry standards. This chapter provides guidance on practicing Cat6 cable wiring using the T568B standard, which is commonly used in networking.

Why Wiring Standards Matter
Wiring standards ensure that cables are consistent and compatible across different devices and network setups. Properly wired cables prevent connectivity issues and ensure reliable data transmission. The T568B standard is widely used in both residential and commercial networking, making it a crucial skill for any network technician.

The T568B Wiring Standard

When wiring a Cat6 cable according to the T568B standard, the wire colors should be arranged as follows:

1. **Pin 1**: White/Orange
2. **Pin 2**: Orange
3. **Pin 3**: White/Green

4. **Pin 4**: Blue
5. **Pin 5**: White/Blue
6. **Pin 6**: Green
7. **Pin 7**: White/Brown
8. **Pin 8**: Brown

Video Tutorial on How to Wire Keystone Jack and How to Replace Network Jack Faceplate

The following is a video of myself wiring and testing a network jack. A narration is included in the video so that viewers may learn and follow along. Included in the video is what to look for when testing a newly wired network cable and what a broken network cable looks like on the Fluke device. And how to replace a faceplate and network jack holders.

video here: How to Wire Network Cable T568B and How to Repair Network Jack

Why Practicing is Important

- **Skill Development:** Regular practice ensures that you can wire cables quickly and accurately, which is essential in real-world networking environments.
- **Error Prevention:** Familiarity with the wiring process reduces the likelihood of mistakes, such as crossing wires or incorrect pin alignment, which can cause connectivity issues.
- **Professional Confidence:** Being proficient in wiring cables according to the T568B standard will give you the confidence to handle similar tasks during your internship and in future roles.

Practice Wiring During Downtime

From my experience as a network operations intern, I found that practicing wiring cables at your desk during downtime matters. I had to rewire several network cables, and it's much easier and more comfortable to practice in your office than in a hot, sweaty, and humid empty dorm room or while fixing a network jack for a student, faculty, or staff member.

Additionally, when you're in the field, you might face time constraints, especially if the person you're helping needs to leave. Struggling with rewiring in those situations can be stressful. That's why it's highly recommended that the person stays in their room while you work so they can test their devices' connection to the network jack immediately after the repair.

Advice on Rewiring

During your internship, you may hear from others that you will mostly be replacing faceplates. While this advice is well-intentioned, I highly recommend not relying solely on it. In my experience, I still had to rewire several cables, so it's important to be prepared and skilled in this area.

And depending on the wire, the orange and brown wire might look very similar so be careful when putting those in the correct pins. Also sometimes the white/orange or white/blue/ or white/brown or white/blue wires might not show the color and only show white so when initially stripping the cable check and make sure what those white wires are already paired with so you know what kind of white wire it is.

Conclusion
Practicing the T568B wiring standard for Cat6 cables is an important step in becoming a proficient network technician. By following the steps outlined in this chapter and dedicating time to practice, you will develop the skills necessary to create reliable network connections and troubleshoot issues effectively. Mastery of this skill will be invaluable as you progress in your internship and future career.

Technical Documentation: Basic Networking Concepts

Title: Technical Documentation for Network Operations Interns

Introduction:

- **Purpose**: This documentation aims to provide new interns with a fundamental understanding of basic networking concepts. It covers key components and terminology essential for anyone involved in network operations.
- **Intended Audience**: New interns at Connecticut College.
- **Overview**: This guide includes definitions, functions, and examples of key networking concepts, including, but not limited to, networks, routers, switches, and common network configurations.

Basic Networking Concepts

What is a Network?

- **Definition**: A network is a collection of computers, servers, mainframes, network devices, or other devices connected to one another to allow the sharing of data.
- **Importance**: Networks enable resource sharing, including internet access, data storage, and applications. They are fundamental to modern communication and business operations.

Types of Networks:

- **Local Area Network (LAN)**:
  - **Description**: A network that spans a relatively small area, typically within a single building or campus.
  - **Example**: The network within Connecticut College's dormitory buildings.
- **Wide Area Network (WAN)**:
  - **Description**: A network that covers a broad area, such as a city, country, or even globally.
  - **Example**: The internet is the largest WAN, connecting computers worldwide.
- **Metropolitan Area Network (MAN)**:
  - **Description**: A network that spans a city or large campus.
  - **Example**: A citywide Wi-Fi network.

Key Network Components:

- **Router**:
  - **Function**: A device that routes data from one network to another based on their IP addresses. It connects different networks, such as a local network to the internet.
- **Switch**:
  - **Function**: A device that connects devices within a single network, allowing them to communicate. It uses MAC addresses to forward data to the correct destination.
- **Modem**:
  - **Function**: A device that modulates and demodulates signals for data transmission over telephone lines, cable systems, or satellite connections.
- **Network Cables**:
  - **Cat5/Cat6**: Types of Ethernet cables used for network connections. Cat6 cables have higher performance specifications than Cat5.
  - **RJ45**: A type of connector used with Ethernet cables.

Understanding Routers

Definition of a Router: A router is a networking device that forwards data packets between computer networks. It routes traffic from one network to another, ensuring data reaches its intended destination.

Functions of a Router:

- **Routing Traffic**: Determines the best path for data packets to travel from the source to the destination.
- **Connecting Networks**: Links different networks, such as a home network to the internet.
- **Managing Traffic**: Uses various protocols to manage and direct network traffic efficiently.

Understanding Switches

Definition of a Switch: A switch is a networking device that connects devices within a network and uses MAC addresses to forward data to the correct destination.

Functions of a Switch:

- **Connecting Devices**: Links multiple devices within the same network, enabling communication between them.
- **Managing Data Traffic**: Directs data packets to their destination based on MAC addresses.

Types of Switches:

- **Managed Switch**:
  - **Description**: Allows for configuration and management, providing greater control over the network.
  - **Use Case**: Used in larger networks where performance and security are priorities.
- **Unmanaged Switch**:
  - **Description**: Plug-and-play with no configuration required, simpler but less flexible.
  - **Use Case**: Used in smaller networks or for basic connectivity needs.

Introduction to Network Jacks:

- **Definition**: Network jacks, typically using RJ45 connectors, are points of access for connecting devices to a network via Ethernet cables.
- **Importance**: Properly functioning network jacks are essential for maintaining stable and reliable network connections.

Common Issues with Network Jacks:

- **Loose Connections**: The Ethernet cable doesn't stay securely in the jack.
- **Damaged Pins**: The metal pins inside the jack are bent or broken.
- **Wiring Problems**: Incorrect or loose wiring inside the jack.
- **Pushed or missing in Network Jacks:** The blue network jack is pushed inside the wall
- **Missing or Broken Parts**: Parts of the whole network box are broken either the face plate, the module holders for the network jacks, or the whole box itself

Repair Tools:

- Fluke Network tester
- Screwdriver set
- Power drill
- Punch-down tool

- Wire stripper
- Wire cutter

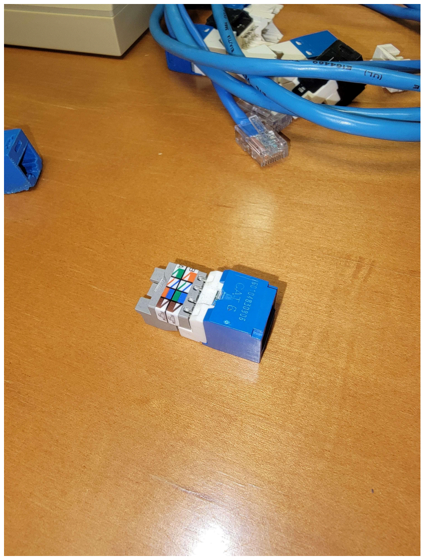
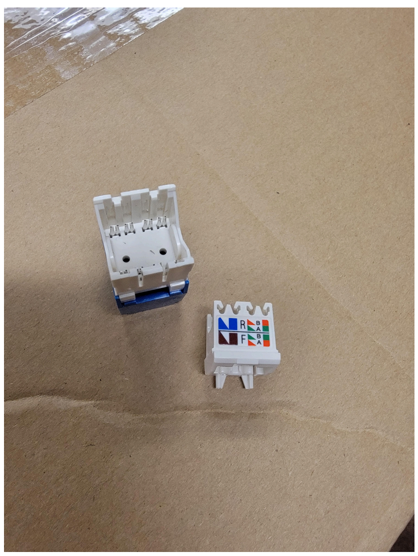
Pictured above are the main tools I used

Step-by-Step Repair Guide

1. **Identify the Issue**:
   - Use an Ethernet cable tester like a Fluke to check if the network jack is functioning properly.
   - Inspect the jack for visible damage, such as bent pins or loose connections.
   - Refer back to the section of examples of network jacks that need to be fixed
   - If you are unsure if it needs to be fixed, take a picture of the network jack and ask a coworker in google chat if it needs to be fixed or not.
2. **Replace a Damaged Jack**:
   - How to Wire Network Cable T568B and How to Repair Network Jack
3. **Test the Repair**:
   - Use the Fluke Network tester to verify that the connection is now working correctly.
   - Check for a stable connection by connecting a Fluke to the network jack and testing internet connectivity.

Cat6 Wires and Cat6 RJ45 Keystone Jack Notes and Resources

- You will be using the following kinds of jacks:
- Keystone Jack Picture

- 
- Mini-Com TX6 Plus Jack Picture
- 
- Regardless of which kind of jack you'll be working with they both still follow the same T568B standards of wiring.

4. If you need to rewire and crimp a network cable and it is already very short either be very careful when cutting the wire to not make it too short or you may need to extend it using an RJ45 coupler and another patch cable. This will add length to the network cable.
   - Here is a picture of a RJ45 coupler
   - 

Understanding VLANs (Virtual Local Area Networks)

What is a VLAN?

A Virtual Local Area Network (VLAN) is a logical grouping of devices on one or more LANs that act as if they are on the same network, even if they are on different physical segments. VLANs improve performance, security, and manageability.

Key Concepts

1. Segmentation: VLANs segment networks based on function, team, or application, reducing the size of broadcast domains for better performance and security.
2. Broadcast Control: VLANs confine broadcast traffic to smaller domains, reducing unnecessary traffic.
3. Security: VLANs isolate sensitive data and resources, enhancing security.
4. Flexibility: VLANs allow devices to be grouped logically, not physically, making network changes easier.

How VLANs Work

- VLAN Tagging: Adds a VLAN ID to network frames, identifying the VLAN the frame belongs to. This is done using the IEEE 802.1Q standard.
- Switch Ports:
    - Access Ports: Assigned to a single VLAN, connecting end devices.
    - Trunk Ports: Carry traffic for multiple VLANs, connecting switches or routers.

Configuring VLANs

1. Assigning Ports: On a managed switch, assign ports to specific VLANs.
2. Inter-VLAN Routing: Use a router or Layer 3 switch to enable communication between VLANs.

Benefits of VLANs

- Improved Performance: Smaller broadcast domains enhance network performance.
- Enhanced Security: Isolation of sensitive data reduces unauthorized access.
- Better Management: Logical segmentation simplifies network management and troubleshooting.
- Increased Flexibility: Devices can be moved within the network without changing physical connections.

Example

A company has three departments: Finance, Marketing, and IT. VLANs are created for each department (e.g., VLAN 10 for Finance). This setup isolates traffic, enhances performance, and simplifies management. If employees move, they stay on the same VLAN without changing cabling.

Finding MAC Addresses on Cisco Devices Using Command Prompts

Overview

This document provides a basic guide to locating and identifying MAC (Media Access Control) addresses on Cisco switches using command-line interface (CLI) commands. MAC addresses are unique identifiers assigned to network interfaces and are crucial for managing and troubleshooting network communications on a physical layer.

Commands for Finding MAC Addresses

1. Viewing the Entire MAC Address Table

Command: show mac address-table
Description: This command displays the complete MAC address table of the switch. It lists all MAC addresses that the switch has learned, along with their associated VLANs and the ports where the devices are connected.

Usage Example:

Switch# show mac address-table

Sample Output:

plaintext
Copy code
```
VLAN    MAC Address       Type        Ports
----    -----------       ----        -----
1       0001.42a8.b7c4    DYNAMIC     Gi0/1
1       0002.3c7f.3a2c    DYNAMIC     Gi0/2
```

2. Finding MAC Addresses on a Specific VLAN

Command: show mac address-table vlan <vlan-id>
Description: This command filters the MAC address table to display only the MAC addresses associated with a specified VLAN. Replace <vlan-id> with the VLAN number you are interested in.

Usage Example:

```
Switch# show mac address-table vlan 10
```

Sample Output:

```
VLAN    MAC Address       Type       Ports
----    -----------       ----       -----
10      0001.42a8.b7c4    DYNAMIC    Gi0/1
10      0002.3c7f.3a2c    DYNAMIC    Gi0/2
```

3. Finding MAC Addresses on a Specific Interface

Command: show mac address-table interface <interface-id>
Description: This command displays the MAC addresses associated with a specific interface on the switch. Replace <interface-id> with the appropriate interface identifier (e.g., Gi0/1).

Usage Example:

```
Switch# show mac address-table interface Gi0/1
```

Sample Output:

```
VLAN    MAC Address       Type       Ports
----    -----------       ----       -----
10      0001.42a8.b7c4    DYNAMIC    Gi0/1
```

Understanding the Output

- **VLAN:** The VLAN ID to which the MAC address is associated.
- **MAC Address:** The unique hardware address of the network interface.

- **Type:** Indicates how the MAC address was learned by the switch:
  - **DYNAMIC:** The MAC address was learned dynamically through network traffic.
  - **STATIC:** The MAC address was manually configured.
- **Ports:** The physical interface on the switch where the device is connected.

Best Practices

- Regularly monitor the MAC address table to ensure that network devices are correctly mapped to the appropriate ports and VLANs.
- Use VLAN-specific queries to troubleshoot connectivity issues within specific segments of the network.
- For large networks, consider exporting the MAC address table for further analysis.

Conclusion

This documentation provides the foundational commands necessary to locate and manage MAC addresses on Cisco switches. Understanding these commands is crucial for effective network management, allowing for precise troubleshooting and ensuring that network devices communicate efficiently.

Understanding the Basics of MAC Addresses

Overview

MAC (Media Access Control) addresses are a fundamental concept in networking, serving as a unique identifier for devices on a network. This chapter provides a basic understanding of MAC addresses, how they function, and their role in network communications.

What is a MAC Address?

A MAC address is a unique 48-bit identifier assigned to the network interface card (NIC) of a device. This address is used to ensure that data packets are delivered to the correct hardware device within a local network. MAC addresses are essential for network operations, particularly at the data link layer (Layer 2) of the OSI model.

Structure of a MAC Address

A MAC address consists of six pairs of hexadecimal digits, separated by colons or hyphens. For example:

00:1A:2B:3C:4D:5E

The structure is as follows:

1. **Organizationally Unique Identifier (OUI):**
   - The first three pairs (24 bits) of a MAC address represent the Organizationally Unique Identifier (OUI). This part of the address is assigned by the IEEE to the manufacturer of the device, ensuring that the first half of the MAC address is unique to the manufacturer.
   - Example: In the MAC address 00:1A:2B:3C:4D:5E, 00:1A:2B is the OUI.
2. **Device Identifier:**
   - The last three pairs (24 bits) are specific to the device and are assigned by the manufacturer. This portion of the MAC address ensures that each device produced by the manufacturer has a unique address.
   - Example: In the MAC address 00:1A:2B:3C:4D:5E, 3C:4D:5E is the device identifier.

Role of MAC Addresses in Networking

1. **Data Transmission:**
   - MAC addresses are used to ensure that data packets reach the correct device on a local network. When a device sends data, the MAC address is used to specify the destination and source within the data packet.
2. **Switching and Routing:**
   - In a network, switches use MAC addresses to forward data to the correct port. When a switch receives a data packet, it reads the destination MAC address and forwards the packet to the appropriate device on the network.
   - Unlike IP addresses, which are used for routing data between networks (Layer 3 of the OSI model), MAC addresses are used for communication within the same local network segment.
3. **Network Security:**
   - MAC addresses can be used in network security settings to restrict access to the network. For example, network administrators can create MAC address filtering rules to allow or deny access to specific devices based on their MAC addresses.

Finding a MAC Address

1. **On Windows:**
   - Open Command Prompt and type ipconfig /all. Look for the "Physical Address" entry under the network adapter details.
2. **On macOS:**

    - Open Terminal and type ifconfig. Look for the "ether" entry under the network interface details.
3. **On Linux:**
    - Open Terminal and type ifconfig or ip addr. Look for the "link/ether" entry under the network interface details.

Conclusion

MAC addresses play a crucial role in ensuring that data reaches the correct devices within a local network. Understanding MAC addresses is fundamental for network troubleshooting, security, and management. This basic knowledge will help you navigate more complex networking concepts as you progress in your internship.

Understanding the Basics of IP Addresses

Overview

IP (Internet Protocol) addresses are essential for identifying devices on a network and ensuring that data is routed correctly between them. This chapter provides an introduction to IP addresses, including their structure, types, and role in network communication, as well as specific details about the IP addresses used in dorm building switches.

What is an IP Address?

An IP address is a unique numerical identifier assigned to each device connected to a network that uses the Internet Protocol for communication. IP addresses allow devices to locate and communicate with each other over a network, whether it's a local network or the broader internet.

Structure of an IP Address

There are two main versions of IP addresses: IPv4 and IPv6.

1. IPv4 (Internet Protocol version 4):
   - An IPv4 address consists of four sets of numbers separated by periods (dots), with each set ranging from 0 to 255. This format is known as "dotted decimal notation."
   - Example: 192.168.1.1
   - An IPv4 address is a 32-bit number, allowing for approximately 4.3 billion unique addresses.
2. IPv6 (Internet Protocol version 6):
   - Due to the limited number of available IPv4 addresses, IPv6 was developed to provide a much larger address space. An IPv6 address consists of eight groups of four hexadecimal digits, separated by colons.
   - Example: 2001:0db8:85a3:0000:0000:8a2e:0370:7334
   - An IPv6 address is a 128-bit number, allowing for a nearly unlimited number of unique addresses.

Types of IP Addresses

1. Public vs. Private IP Addresses:
   - Public IP Address: A public IP address is assigned to a device connected directly to the internet. This address is unique across the entire internet, allowing external devices to communicate with it.
   - Private IP Address: A private IP address is used within a local network and is not accessible directly from the internet. Private IP addresses are used to identify devices within the same network.

- Example of Private IP Addresses in Dorm Buildings:
    - In the dorm building switches, all IP addresses are private and start with 10.255.255. This range is part of the private IP address space reserved for internal network use, ensuring that these addresses are not routable on the public internet.

2. Static vs. Dynamic IP Addresses:
    - Static IP Address: A static IP address is manually assigned to a device and does not change over time. Static IP addresses are often used for servers, printers, and other devices that require a consistent address.
    - Dynamic IP Address: A dynamic IP address is automatically assigned by a DHCP (Dynamic Host Configuration Protocol) server and can change over time. Most devices on a network, such as computers and smartphones, use dynamic IP addresses.

Role of IP Addresses in Networking

1. Routing Data:
    - IP addresses are used to route data packets between devices on different networks. When a device sends data, the IP address specifies the destination device, allowing routers to forward the data to the correct network.
2. Subnetting:
    - IP addresses are often divided into subnets to create smaller, more manageable networks within a larger network. Subnetting allows for more efficient use of IP addresses and helps organize the network for better performance and security.
3. Network Address Translation (NAT):
    - NAT is a technique used to map multiple private IP addresses to a single public IP address. This is commonly used in home and office networks to allow multiple devices to share a single internet connection.

Finding an IP Address

1. On Windows:
    - Open Command Prompt and type ipconfig. Look for the "IPv4 Address" entry under the network adapter details.
2. On macOS:
    - Open Terminal and type ifconfig. Look for the "inet" entry under the network interface details.
3. On Linux:
    - Open Terminal and type ifconfig or ip addr. Look for the "inet" entry under the network interface details.

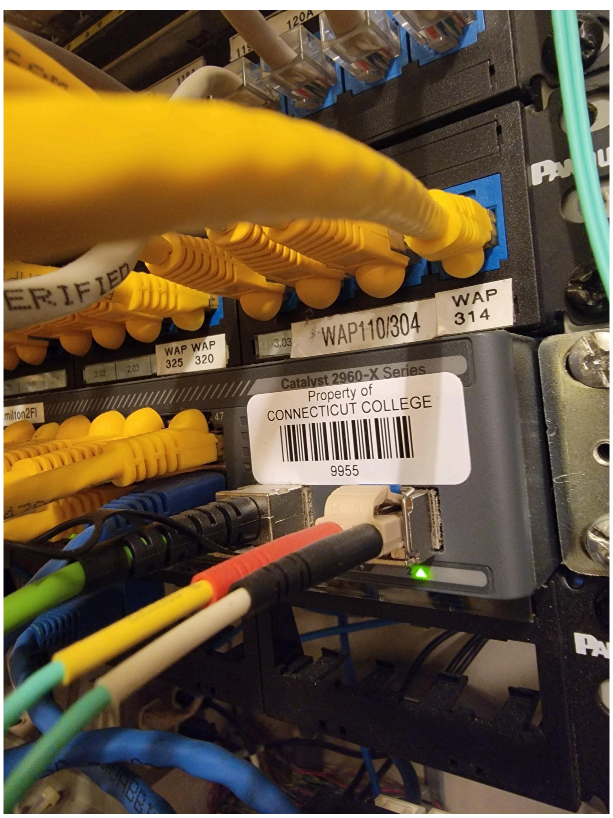
Conclusion
IP addresses are fundamental to the operation of networks, enabling devices to communicate and data to be routed correctly. Understanding the basics of IP addresses, including their structure, role in networking, and specific uses within your environment (such as the private IP addresses in dorm building switches), is crucial for anyone working in network operations. This knowledge will serve as a foundation for more advanced networking concepts.

Dorm Building Switch Closet Locations

As you go through the dorm buildings and check each room to see if there are repairs that need to be made or not you may also be tasked with finding and recording the asset tags of each switch in the building and sending them back to Brandon for him to log. Some switches might not have asset tags. Alert either Brandon or in the NSS chat about this.

Please note you may not be tasked with this as an extra assignment. Your internship may differ from one intern to another.

Asset tags look like this:



Most switches will be behind doors that say “Data Room” but a few are not. Some are behind doors with no signs or signs that don’t indicate there being a switch behind them.

Thoroughly checking each door on the floor where the switch is located is recommended.

Be aware that some dorm buildings contain multiple switches spread across different floors.

<u>The location of the switch closets within each dorm building are listed as follows:</u>

Jane Adams First Floor

Jane Adams Third Floor

Blackstone Basement

Branford Basement

Smith Basement

Burdick Second Floor

Freeman Basement

Freeman Third Floor

Hamilton Basement

Hamilton Second Floor

Johnson First Floor

Katherine Blunt Basement

Katherine Blunt Third Floor

Knowlton Basement: note go through a second door after entering first door to find switch.

Lambdin Basement

Larrabee First Floor

Lazrus Basement

Manwaring Apts Basement

Morrison Basement

Park Second Floor

Plant Basement

River Ridge Apt 11

Windham Basement

Wright Basement

Abbey House Basement

Horizon House Basement

Assigning new cable to correct VLAN on cisco switch

Log into the Switch:

- Connect to the switch using SSH or console.
- Use the command: ssh username@switch_IP_address.
- Enter password when prompted: ask coworkers

Enter Global Configuration Mode:

- Type configure terminal or conf t.

Identify the Interface:

- Determine which port the cable is plugged into. For example, if the cable is plugged into port 1 of the switch, the interface will be Gix/x/x.
- Use the command show interface status to confirm the port status and connected device.

Assign the Interface to a VLAN:
Enter the interface configuration mode by typing:

interface Gix/x/x

To assign the interface to a specific VLAN, use the following commands:

switchport mode access
switchport access vlan X

- Replace X with the VLAN ID you want to assign the port to.

Exit and Save Configuration:

- Type exit to leave the interface configuration mode.

To save the changes, use the command:

write memory

- Alternatively, you can use end followed by copy running-config startup-config to save.

Verify the VLAN Assignment:
To verify that the interface has been assigned to the correct VLAN, use:

show vlan brief

- This command will display all VLANs and their associated ports.

Possible Tickets You May Be Assigned and My Insights and What I learned From Them

Purpose of this Section:
This section of my guide is to show the lessons I've learned while out in the field attending to Web Help Desk tickets I was assigned.

Resolving Apple TV Connectivity Issues Due to Switch Malfunction

Overview
During your internship, you may encounter devices that are not connecting to the internet despite being physically connected via an Ethernet cable. This chapter details a specific case involving an Apple TV that was not connecting to the internet, the troubleshooting steps taken, and how the issue was resolved by identifying a malfunctioning network switch.

Issue Description
A support ticket was assigned to me for an Apple TV that could not connect to the internet. The device was connected via an Ethernet cable, but it was still unable to establish a network connection.

Troubleshooting Steps

1. **Initial Verification:**
    - Confirmed that the Ethernet cable was properly connected to both the Apple TV and the network jack.
    - Verified that the Apple TV's network settings were configured correctly.
2. **Switch Closet Investigation:**
    - Suspected that the issue might be related to the network switch to which the Apple TV was connected.
    - Accessed the switch closet to identify the specific switch and port the Apple TV was plugged into.
3. **Switch Testing:**
    - Tested the switch using a Fluke network tester to verify its functionality.
    - Disconnected the Ethernet cable from the Apple TV and plugged it into a laptop. The laptop also failed to connect to the internet, confirming that the issue was not isolated to the Apple TV.
4. **Switch Diagnosis:**
    - Determined that the switch was not providing internet access to any connected devices.
    - Tested multiple ports on the switch, all of which failed to provide a network connection.
5. **Resolution:**
    - Moved the Ethernet cable connected to the Apple TV to a different, functioning switch within the switch closet.
    - Reconnected the Apple TV and verified that it successfully connected to the internet.
6. **Further Testing:**
    - Used the Fluke tester again to confirm that the new switch was providing proper network access.
    - Verified that other devices connected to the new switch were also able to access the internet without issues.

Lessons Learned

- **Importance of Testing Switches:** This case highlights the importance of thoroughly testing network switches when diagnosing connectivity issues. A malfunctioning switch can affect multiple devices, so it's crucial to verify the switch's performance using tools like a Fluke tester.
- **Understanding Network Layouts:** Familiarize yourself with the network layout, including the location of switches and how devices are connected. This knowledge will help you quickly identify potential points of failure.

- **Documentation:** Always document the troubleshooting process and the steps taken to resolve the issue. This ensures that others can reference your work and understand the solution if similar issues arise in the future.

Best Practices for Similar Issues

- **Systematic Testing:** When a device cannot connect to the network, always test the connection using another device to rule out device-specific issues.
- **Switch Isolation:** If multiple devices are affected, isolate the issue by testing the switch directly. This can help you quickly determine whether the switch or the cabling is at fault.
- **Fluke Tester Utilization:** Use a Fluke tester or similar tool to assess network performance at different points, such as the switch, cabling, and device connection, to pinpoint the source of the problem.

Conclusion
Successfully resolving connectivity issues like the one described above requires a methodical approach and the use of appropriate diagnostic tools. By understanding how to test and diagnose network switches, you can efficiently address similar problems and ensure consistent network performance for all connected devices.

<u>Installing a Cisco Phone and Resolving Power over Ethernet (PoE) Issues</u>

Overview
During your internship, you may be tasked with installing new Cisco phones for employees. A common issue you might encounter is a phone not turning on after installation due to a lack of Power over Ethernet (PoE). This chapter will guide you through the troubleshooting and resolution process, including how to identify and manage PoE ports on a switch.

Issue Description
You were assigned to install a new Cisco phone in an employee's office. After connecting the phone to the network jack, the phone did not power on, indicating that it was not receiving PoE, which is required for the phone to function.

Troubleshooting Steps

1. **Initial Installation:**
   - Connected the Cisco phone to the network jack in the employee's office using the provided Ethernet cable.
   - Verified that the phone was properly connected, but the phone did not power on, indicating it was not receiving PoE.
2. **Switch Closet Investigation:**
   - Accessed the switch closet in the building to check the network configuration.

    - Located the room number on the patch panel to identify which port on the switch the network jack was connected to.
3. **Checking for PoE Availability:**
    - Determined that the phone was connected to a switch that did not support PoE.
    - Moved the cable for the room from the non-PoE switch to a switch that had PoE capabilities.
4. **Managing PoE Ports:**
    - Discovered that all PoE ports on the switch were already in use, preventing the phone from receiving power.
5. **SSH Login and Port Management:**

Logged into the switch using SSH with the command:

ssh conncisco@[IP address of the switch]

-

Used the following command to check the status of the ports and determine which ones were using PoE:

show power inline

-
- Identified ports that were not actively using PoE but were connected to devices.

6. **Reconfiguring Port Connections:**
    - Moved the cables for the devices not requiring PoE from the PoE switch to the non-PoE switch, freeing up PoE ports.
    - Reconnected the cable for the Cisco phone to the now available PoE port.
7. **Verification:**
    - After moving the necessary cables, the Cisco phone successfully powered on and connected to the network.
    - Verified that all other devices connected to both the PoE and non-PoE switches were functioning correctly.

Lessons Learned

- **PoE Port Management:** Managing PoE ports effectively is crucial, especially when dealing with limited resources. Understanding which devices require PoE and which do not can help optimize the use of available ports.

- **Utilizing SSH for Network Management:** SSH is a powerful tool for remotely managing switches, allowing you to check port status, configure settings, and troubleshoot issues without physically interacting with the switch.
- **Flexibility in Problem Solving:** When faced with limitations, such as a full PoE switch, it's important to be flexible and think creatively. In this case, reallocating non-PoE devices to another switch was an effective solution.

Best Practices for Installing Cisco Phones

- **Check PoE Requirements:** Before installing a Cisco phone, ensure that the network jack it will connect to is linked to a PoE-capable switch.
- **Verify Port Usage:** Regularly monitor port usage on PoE switches to ensure that critical devices like phones have the power they need.
- **Document Changes:** Always document any changes made to switch configurations, including which ports were moved, to maintain a clear record for future reference.

Conclusion
Successfully installing network devices like Cisco phones often requires more than just physical connections. Understanding how to manage network switches and PoE resources is key to ensuring that all devices function properly. This chapter outlines the importance of being proactive in managing network resources and the steps needed to resolve issues that may arise during installation.

<u>Fixing a Student's Network Jack and the Importance of Device Testing</u>

Overview
When addressing network issues in student dorm rooms, it's crucial to ensure that the problem is fully resolved not just from a technical standpoint, but also in terms of the end-user experience. This chapter discusses a scenario where I repaired a student's network jack, the importance of having the user present during the repair, and the lessons I learned about ensuring device-specific connectivity.

Issue Description
I was assigned a ticket to repair a network jack in a student's dorm room. Upon arrival, the student asked if they needed to stay in the room while I performed the repair. I informed them that it wasn't necessary, and they left before I began the work.

Steps Taken

1. **Initial Interaction:**

    - The student asked if they needed to remain in the room during the repair. I advised them that it wasn't necessary, and they left the room.
2. **Repair Process:**
    - I proceeded to diagnose the issue with the network jack.
    - Using the Fluke device, I tested the connection and identified the problem with the network jack.
    - I repaired the network jack and retested it using the Fluke device to ensure that it was providing internet access.
3. **Realization After Repair:**
    - After completing the repair, I realized that having the student present to test their own devices would have been beneficial. While the Fluke device confirmed the jack was providing internet access, it would have been ideal to have the student test their specific device to ensure it was receiving the connection properly.
4. **Follow-Up Actions:**
    - After leaving the room, I went to Web Help Desk and sent an email to the student. In the email, I informed them that I had fixed their network jack and tested it with my network device. I also asked them to test their device to make sure it was working properly.
5. **Student's Response:**
    - The student later emailed me back, confirming that their device was working after I had repaired the network jack. While everything worked out, I realized it would have been better if I had asked them to stay during the repair to ensure their device was connected to the internet immediately.

Lessons Learned

- **End-User Testing is Crucial:** While testing the network jack with a Fluke device confirmed that it was providing internet access, I learned that it's important to remember that the end-user's device might have specific settings or issues that could affect connectivity. Ensuring that the user is present to test their own device helps confirm that the issue is fully resolved.
- **Communication with Users:** When users asked if they needed to stay during a repair, I realized it's beneficial to explain why their presence might be necessary. Communicating that testing their specific device is an important part of the repair process can prevent potential issues and ensure complete satisfaction.
- **Follow-Up if User is Absent:** If the user was not present, send a follow-up message in web help desk advising them to test their device and update if they experienced any issues.

Best Practices for Future Repairs

- **Request User Presence:** In the future, I would politely request that the user remain in the room during repairs to test their device immediately after the repair is completed. I would explain that this ensures the issue is fully resolved for their specific setup.
- **Perform Thorough Testing:** Even if the user was not present, I used the Fluke device and any other available tools to ensure that the network jack was providing internet access. If possible, I would connect a standard device like a laptop to double-check connectivity.
- **Follow-Up Communication:** If the user was absent, I would leave clear instructions for them via Web Help Desk or email to test their device as soon as possible and to report any issues immediately.

Conclusion

Ensuring that a network jack was functioning was more than just confirming that it provided internet access. It was also about making sure that the end-user's specific device could connect to the network without issues. This chapter highlights the importance of having the user test their own device and the value of clear communication to achieve complete resolution of network issues.

<u>Handling Tickets Assigned on Behalf of Another Client</u>

Overview

In some instances, you may be assigned tickets where the person who submitted the request is not the one experiencing the issue but is submitting it on behalf of someone else. This chapter discusses how to handle such situations, including the importance of identifying and contacting the correct client, and how to use the CamelWeb directory for accurate communication.

Issue Description

There were times when I was assigned tickets where the client who submitted the request was not the actual person in need of repairs. Instead, they were submitting the ticket on behalf of someone else, such as a colleague or a student.

Steps Taken

1. **Identifying the Real Client:**
    - Upon receiving a ticket, I first checked if the person who submitted the ticket was the one needing the repair. If it was indicated that they were submitting on behalf of another person, I needed to identify the actual client.
2. **Using the CamelWeb Directory:**
    - To ensure proper communication and to avoid delays, I used the CamelWeb directory to look up the contact information for the actual client.

    - CamelWeb provided detailed contact information, including email addresses and phone numbers, which allowed me to reach out directly to the person experiencing the issue.
3. **Contacting the Correct Client:**
    - After identifying the correct client, I contacted them to discuss the issue, confirm details, and schedule a time for the repair.
    - This step ensured that I was working directly with the person who needed the repair, which helped streamline the process and avoid miscommunications.

Lessons Learned

- **Accurate Communication is Key:** When handling tickets submitted on behalf of someone else, it's crucial to establish direct communication with the person who actually needs the service. This helps prevent misunderstandings and ensures that the issue is addressed correctly.
- **Using Available Resources:** The CamelWeb directory is an essential tool for locating and contacting the correct client. Familiarizing yourself with this resource can save time and improve the efficiency of your work.
- **Follow-Up is Important:** After completing the repair, I made sure to follow up with the correct client to confirm that the issue was resolved to their satisfaction. This also provided an opportunity for them to test the repair immediately, ensuring that everything was functioning as expected.

Best Practices for Handling Third-Party Tickets

- **Always Verify the Client:** If a ticket is submitted on behalf of someone else, verify who the actual client is before proceeding with the repair.
- **Use CamelWeb Effectively:** Leverage the CamelWeb directory to quickly find contact information and ensure you're communicating with the right person.
- **Clear Communication:** When contacting the correct client, explain the situation and confirm the details of the issue before scheduling any repairs.

Conclusion
Handling tickets submitted on behalf of others requires careful attention to detail and clear communication. By using resources like the CamelWeb directory and ensuring you are in contact with the right person, you can effectively manage these situations and provide the necessary support to resolve the issue.

Assigning a Switch Port to the Correct VLAN for Internet Access

Overview
Ensuring that a computer has proper internet access involves more than just physically connecting it to the network. This chapter details a situation where I was assigned to provide internet access to a computer, the steps I took, and how I resolved the issue by configuring the correct VLAN for the switch port.

Issue Description
I was assigned a ticket to provide internet access to a computer. After physically connecting the Ethernet cable to both the computer and a network switch, I discovered that the computer was still unable to access the internet.

Steps Taken

1. **Initial Setup:**
   - I connected the Ethernet cable to the computer and ensured that the other end was properly plugged into a network switch.
   - Despite this connection, the computer was not able to access the internet.
2. **Troubleshooting the Network Connection:**
   - I verified that the Ethernet cable was functioning correctly and that it was securely connected to the appropriate switch port.
   - I then realized that the issue might be related to the network configuration rather than the physical connection.
3. **Accessing the Switch via SSH:**
   - I logged into the switch using SSH to check the configuration of the port to which the computer was connected.

Command used to log in:

ssh conncisco@[IP address of the switch]

Password - ask coworkers

4. **Assigning the Correct VLAN:**
   - Upon reviewing the switch configuration, I found that the port was not assigned to the correct VLAN, which was necessary for the computer to access the academic network.

I assigned the port to the correct academic VLAN using the following commands:
bash

interface [interface-id]

```
switchport mode access

switchport access vlan [vlan-id]
```

    ○ This ensured that the computer was now connected to the appropriate VLAN that provided internet access.

5. **Verification:**
    ○ After making the changes, I confirmed that the computer was able to access the internet by testing the connection.
    ○ The computer successfully connected to the internet, resolving the issue.

Lessons Learned

- **VLAN Configuration is Critical:** Assigning the correct VLAN is essential for ensuring that devices are connected to the appropriate network segments. Without this step, even properly connected devices may not be able to access network resources.
- **Thorough Troubleshooting:** When a device isn't connecting to the internet, it's important to consider both physical and logical network configurations. Checking VLAN assignments can often resolve issues that aren't immediately obvious.
- **Using SSH for Configuration:** SSH is a powerful tool for accessing and configuring network switches remotely. Familiarity with command-line tools and network configurations is crucial for troubleshooting and resolving network issues.

Best Practices for Providing Internet Access

- **Check Physical Connections First:** Always verify that cables are securely connected and functioning before moving on to network configuration checks.
- **Verify VLAN Assignments:** If a device isn't connecting to the internet, check the switch port's VLAN configuration to ensure it's assigned to the correct network segment.
- **Test After Changes:** After making any network configuration changes, always test the device to ensure the issue has been resolved.

Conclusion
Providing internet access to a computer involves both physical and network configuration steps. This chapter highlights the importance of VLAN assignments and thorough troubleshooting to ensure that devices are properly connected to the network. By following these steps, you can effectively resolve similar issues in the future.

<u>Handling Network Jack Repairs When the Client is Absent</u>

Overview
There will be times during your internship when you'll need to perform a network jack repair

while the person is not present in their dorm or office. In these situations, it's crucial to ensure the repair is thoroughly tested and that the client is informed of the work done.

Procedure

1. **Performing the Repair:**
   - Begin by diagnosing and repairing the network jack as you would normally.
   - Use the Fluke device to test the network jack and confirm that it is providing proper internet access.
2. **Testing the Connection:**
   - Ensure that the Fluke device indicates a successful connection. If possible, connect a standard device like a laptop to double-check the network jack's functionality.
3. **Follow-Up Communication:**
   - Since the client isn’t present to test their own device, it's important to follow up with them. Send an email through Web Help Desk informing the client that the repair has been completed and the network jack has been tested.
   - In the email, request that the client tests their device as soon as possible to confirm that the network jack is functioning correctly.
4. **Monitor for Feedback:**
   - After sending the follow-up email, monitor for any responses from the client. If they report issues, be prepared to revisit the site and address any further problems.

Best Practices

- **Always Test with the Fluke Device:** This ensures that the network jack is providing the necessary connectivity, even if the client is not present to test it themselves.
- **Clear Communication:** Keep the client informed about the repair and provide them with instructions to test their own equipment.
- **Prompt Follow-Up:** Make sure to follow up with the client soon after the repair to ensure they have a seamless experience.

Conclusion
Handling network jack repairs when the client is absent requires careful attention to detail and proactive communication. By thoroughly testing the network jack and following up with the client, you can ensure that the repair is successful and that the client is satisfied with the service.

Troubleshooting a Persistent Network Jack Issue

Overview
During your time as a network operations intern, you may encounter network issues that are difficult to resolve, requiring a thorough troubleshooting process and, in some cases, the

escalation of the issue to another department. This chapter outlines the steps I took when faced with a persistent network jack problem that ultimately needed to be escalated due to what appeared to be an internal cabling issue.

Issue Description

I received a ticket about a network jack in a dorm room that was not providing a connection. When I initially tested the jack using a Fluke network device, it displayed the message "connected to unpowered device," indicating that the network jack was not functioning as expected.

Troubleshooting Steps

1. **Initial Testing with Fluke:**
    - I plugged the Fluke device into the network jack to test for connectivity. The Fluke device reported "connected to unpowered device," which typically suggests that the jack is connected to something that isn't providing power or network connectivity.
2. **Replacing the Network Jack:**
    - To rule out the possibility of a faulty network jack, I replaced the network jack three times. After each replacement, I tested the connection again with the Fluke device, but the result remained the same—"connected to unpowered device."
3. **Checking the Switch Patch Panel:**
    - Next, I checked the connection at the switch patch panel. I swapped the port on the patch panel that the network jack was connected to, but this did not resolve the issue. The Fluke device continued to indicate the same problem.
4. **Tracing the Cable with a Network Tone Device:**
    - I used a network tone device to trace the correct cable that was connected to the network jack in the dorm room. I successfully identified the cable by following the tone, confirming that it was the correct one connected to the jack.
    - Despite confirming the cable's path and connection, the network jack still showed no signs of improvement.
5. **Testing the Switch Port:**
    - I also tested the port on the switch to which the patch panel was connected. The port was functioning correctly, which further ruled out issues with the switch or patch panel.
6. **Conclusion and Escalation:**
    - After exhausting all other troubleshooting steps, I concluded that the issue was likely due to the cabling inside the wall, which could not be easily accessed or repaired by our department.

- Given the persistent nature of the problem and the unavailability of a clear solution, I escalated the ticket to an adjacent department that specializes in internal cabling and infrastructure repairs. They would be better equipped to investigate and resolve the issue.

Lessons Learned

- **Thorough Troubleshooting:** This experience underscored the importance of systematically testing and ruling out all possible issues before concluding that a problem is beyond your ability to resolve.
- **Knowing When to Escalate:** It's essential to recognize when an issue needs to be escalated to a specialized team. Doing so ensures that the problem is handled by those with the appropriate expertise and tools.
- **Documentation:** Documenting the steps taken and the results obtained is crucial when escalating an issue, as it provides the next team with a clear understanding of what has been done so far.

Conclusion
While not all network issues can be resolved at the first level of support, the key is to methodically troubleshoot the problem, document your findings, and escalate when necessary. This approach ensures that network issues are resolved as efficiently as possible, with minimal disruption to users.

The Importance of Attention to Detail

Overview

In network operations and IT support, attention to detail is crucial for ensuring that tasks are completed correctly the first time, minimizing errors, and preventing future issues for users. Overlooking small details can lead to larger problems that require revisiting a task, troubleshooting, or even escalating the issue. In this chapter, I'll highlight two instances from my internship that taught me the importance of paying close attention to every step in the process.

Example 1: Cisco 7841 Phone Handset Installation

During one of my tasks, I was responsible for installing a **Cisco 7841 phone** for a client. I successfully completed the installation and tested the phone, but soon after leaving, I received an email from the client stating that they could receive calls, but the person on the other end couldn't hear them speaking. This indicated that there was an issue with the microphone or handset.

When I returned to troubleshoot, I realized I had plugged the phone's **handset cable** into the port designated for **headsets** rather than the port for the handset. The ports were positioned right next to each other with only a small symbol to distinguish them, and I hadn't noticed the difference during the initial installation.

Lesson Learned: This experience emphasized how crucial it is to pay attention to even the smallest details, like the markings on ports and connections. Taking an extra moment to double-check connections or ports could have saved time and avoided the need for troubleshooting. Ensuring the correct port is used for the right device can be the difference between a smooth setup and a follow-up problem for the client.

Example 2: Troubleshooting Call Forwarding Settings on a Cisco Phone

In another instance, I was troubleshooting a phone that wasn't working even though it was plugged in correctly. I accessed the **Cisco admin web portal** to check the device's settings and noticed that the call forwarding settings were incorrect. I made the necessary adjustments to the correct call forwarding option.

However, when I changed the call forwarding settings, I hadn't realized that other settings were also affected by this change. As a result, even though I corrected the call forwarding, the phone still wasn't functioning as expected because other critical settings had been altered.

I returned to investigate further and realized that by changing one option, other settings had automatically been modified, and I hadn't reviewed all the changes. After fixing the additional settings, the phone worked correctly.

Lesson Learned: This situation taught me that when making changes to a system or device, it's important to be aware of all the potential impacts of those changes. Often, modifying one setting can affect other interconnected settings or functions. Reviewing everything carefully before and after changes ensures that nothing is overlooked.

Conclusion

These experiences reinforced the importance of attention to detail in my work. Small oversights, such as plugging a cable into the wrong port or not reviewing all changes after adjusting a setting, can lead to additional issues that require more time and effort to resolve. By being more mindful of these details and double-checking my work, I can ensure smoother operations and avoid unnecessary problems for both myself and the end users.

Attention to detail is not just about avoiding mistakes—it's about understanding the broader impact of every action, ensuring thoroughness in your work, and maintaining a higher standard of performance.

The Importance of having the Client Present When doing a Ticket

Overview

In network operations and IT support, communication with the client is key to ensuring that issues are resolved effectively and that expectations are met. One important aspect of this communication is having the client present during repairs or troubleshooting. This chapter will explain why having the client present can make a significant difference, based on an experience I had with a network jack repair. However, it's also important to recognize when the client's presence may not be necessary, as some tasks are simple enough to resolve without them being there.

Example 1: Network Jack Repair in a Dorm Room

I received a ticket to repair a network jack in a student's dorm room. In the messages exchanged with the client, he mentioned that the network jack was working, but it wasn't fitting correctly into the faceplate. He also said I could do the repair while he was out of his room, which I thought would be a quick and straightforward task.

When I arrived at the dorm, I realized that this was the same dorm room where I had performed a network jack repair about a month ago. Upon inspecting the network jack, it looked exactly the same as it did after my original repair, and there didn't seem to be any real issue with its functionality or placement. To ensure there wasn't anything I had missed, I took a picture and shared it with my coworkers in the **Network Support Services (NSS)** chat. They confirmed that the jack looked fine and that there was no apparent problem.

The Problem with Not Having the Client Present
Had the client been present during the repair, it would have been much easier to explain in person why the network jack was functioning and appeared as it should. Instead, I had to follow up with the client over email to explain that there wasn't a need for further repairs and that the jack was in good condition.

When explaining technical details like this via email, it can be more difficult to clearly communicate the nuances of the repair or why something is functioning as expected. Additionally, if the client has follow-up questions or concerns, those can be addressed more effectively in person, where you can show them the work directly rather than through photos or messages.

Example 2: Simple Phone Reset

Not all tickets require the client to be present. In some cases, the issue is easy to resolve, and it's not necessary to have the person there during the repair. For example, I received a ticket for a phone that was not working. When I arrived at the client's office, they were not present, but the issue turned out to be very simple. I resolved the problem by simply unplugging the phone and plugging it back in, which restored its functionality.

In such cases, having the client present would not have added any value, and the issue could be fixed quickly without the need for further communication.

Using Your Best Judgment

While it's generally beneficial to have the client present during more involved repairs or troubleshooting, not all tickets will require this. Use your best judgment to determine whether the client's presence is necessary. Consider the complexity of the issue and whether the client needs to test the functionality of their device before you leave. For straightforward fixes, such as resetting a device, you can often complete the task without the client being there.

Benefits of Having the Client Present

1. **Immediate Feedback and Confirmation:**
    - When the client is present during the repair, they can test their devices and confirm that everything is functioning correctly on the spot. This avoids the need for follow-up communication and ensures both parties are satisfied before leaving the job site.
2. **Clearer Communication:**
    - In person, it's easier to explain the technical details of what has been done or why no further work is required. Visual demonstrations or hands-on examples can be used to clarify any confusion the client might have.
3. **Preventing Future Tickets:**
    - If the client is not present and doesn't fully understand the repair, there's a chance they may submit another ticket for the same issue in the future, thinking the problem still persists. Having them there ensures they fully understand the resolution and reduces the likelihood of repeat tickets.
4. **Addressing Misunderstandings Quickly:**

    - In this particular case, the client believed the network jack wasn't fitting properly, but had he been there, I could have shown him that it was actually installed correctly. This would have saved time and eliminated any further back-and-forth communication.

Conclusion

Having the client present during repairs, even for what seem to be straightforward tasks, can greatly improve communication and the efficiency of resolving tickets. Not only does it allow for immediate confirmation of the work done, but it also helps prevent misunderstandings that might arise from remote communication. However, for simpler issues, like a phone reset or straightforward network check, using your best judgment to decide whether the client needs to be there can save time. This balance will help you manage tickets efficiently while ensuring client satisfaction.

Using Cisco Unified CM Admin to Resolve Call Forwarding Requests

In this section, I'll explain how I helped resolve a ticket involving the setup of a call forwarding system for a client and their coworker using Cisco Unified CM Admin. I worked alongside a coworker who guided me through the process, as this ticket involved advanced configuration.

Ticket Overview:

The client had a new hire and wanted their phone system set up so that:

1. When a call comes in and neither the client nor their coworker picks up after 3 rings, the call forwards to the new hire.
2. If the new hire also doesn't pick up, the call returns to either the client or their coworker, and the caller leaves a voicemail in the voicemail box of the person they originally tried to reach.

To achieve this, we used Cisco Unified CM Admin to configure call forwarding rules and manage the hunt pilot, hunt list, and line group.

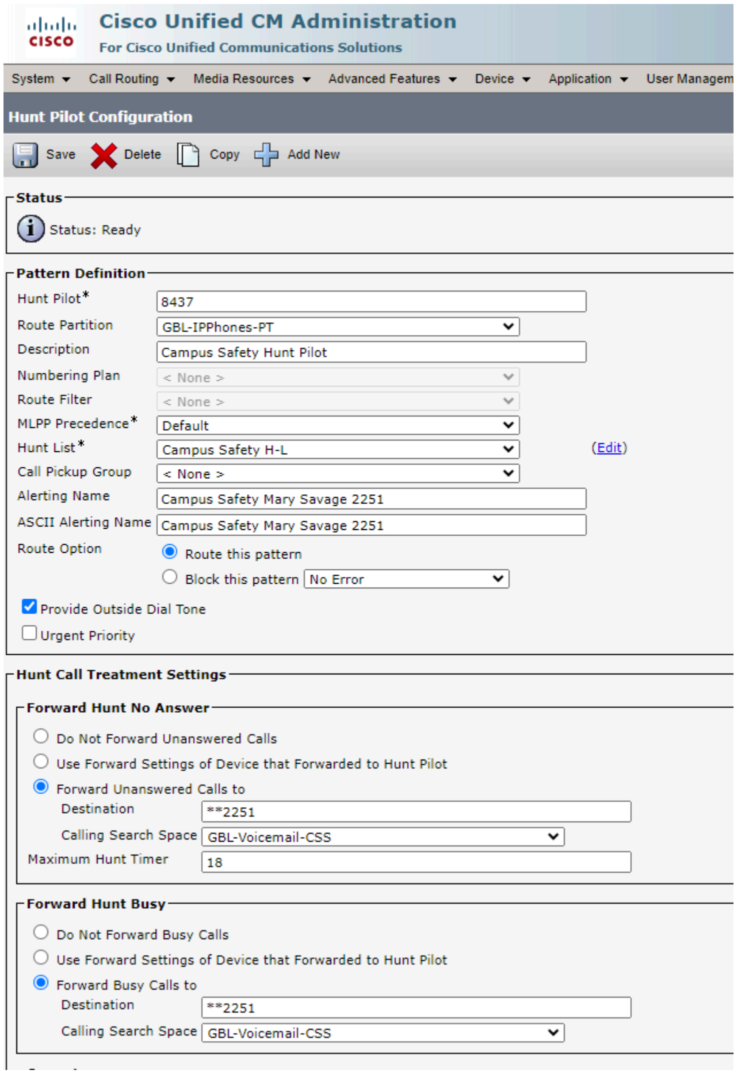
Step-by-Step Solution:

1. Creating Hunt Pilots for the Client and Coworker

We began by creating Hunt Pilots in Cisco Unified CM Admin for both the client and their coworker. Hunt Pilots are used to direct incoming calls to the appropriate extensions based on predefined rules. Here's how we set this up:

- Each of the hunt pilots was assigned their own extension. These extensions are essentially the numbers that the phones of the client and their coworker are tied to.
-

- Pictured here:



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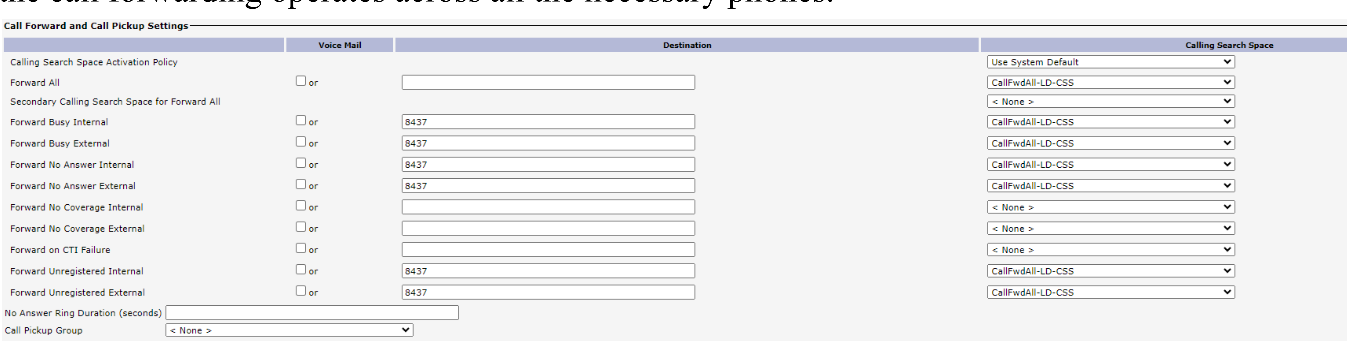
- In the Call Forward Settings for the hunt pilots, we entered the extension for the coworker.
- The Calling Search Space (CSS) was set to And in the Directory Number Configuration in the Call Forward and Call Pickup Settings in the Destination section we put the Hunt Pilot's extension for the client and then in the Calling Search Space setting was set to CallFwdAll-LD-CSS, which defines the range of numbers that can be dialed and ensures the call forwarding operates across all the necessary phones.



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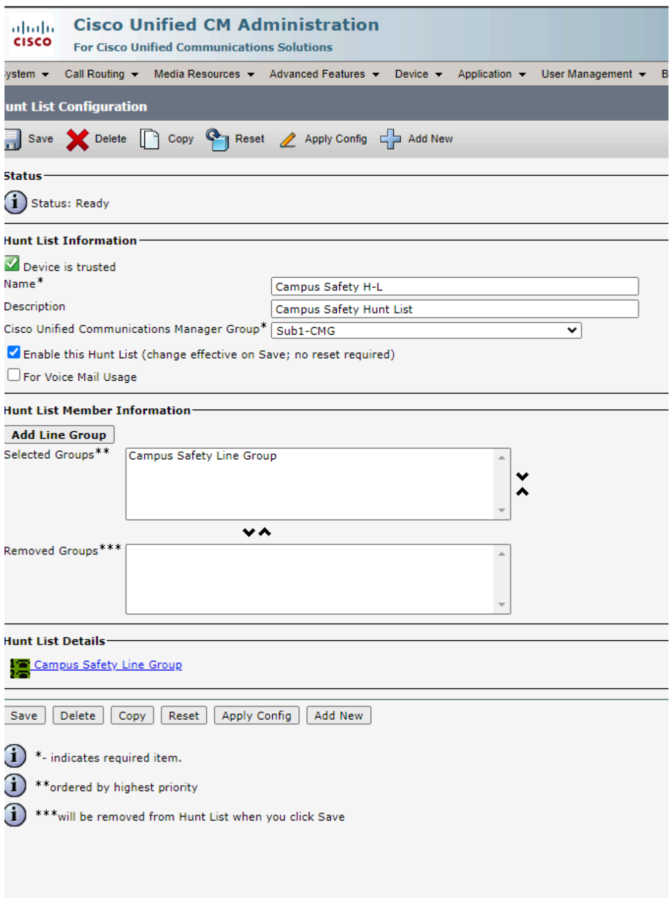
2. Creating a Hunt List and Line Group

Next, we needed to create a Hunt List to manage the sequence of call forwarding. The hunt list ensures that if the initial recipients (client or coworker) don't answer, the call follows a defined path. Here's what was done:

- A Hunt List was created within the system. This hunt list contains the sequence of lines that the system will call if the original recipients don't answer.



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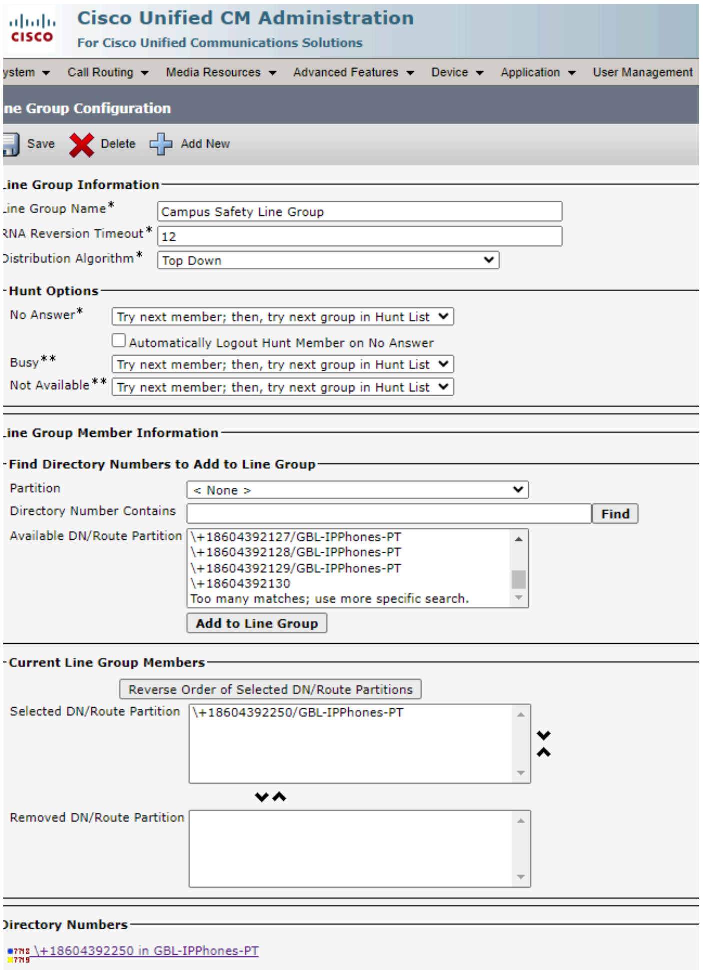
- Within the hunt list, a Line Group was created. The line group included the new hire's extension. This means that if the client or their coworker didn't pick up after 3 rings, the system would forward the call to the new hire.



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3. Configuring the Hunt Pilot to Redirect Calls

Once the hunt pilots, hunt list, and line group were set up, the final step was to configure the behavior of the hunt pilot:

- The Hunt Pilot settings were configured to redirect the call to the new hire if the client or their coworker didn't pick up after 3 rings. This ensured that the new hire would be next in line to answer the call.
- If the new hire also didn't answer, the call was redirected back to either the client or their coworker. The system would then prompt the caller to leave a voicemail in the voicemail box of the person they originally tried to reach.

Using a Fluke Network Device

Overview

The Fluke Network Device is an essential tool for network technicians, providing a wide range of functionalities that assist in diagnosing and troubleshooting network issues. This chapter will guide you through the basic use of the Fluke device, including how to test network jacks, identify VLAN assignments, and verify cable integrity.

Understanding the Fluke Network Device

The Fluke device is versatile and can be used for various tasks, such as cable testing, network verification, and VLAN identification. It's equipped with several options and modes, each designed to assist in different aspects of network management.

Steps to Use the Fluke Network Device

1. **Power On the Device:**
    - Begin by powering on the Fluke device. Ensure it has sufficient battery life for the tasks you plan to perform.
    - Familiarize yourself with the device's interface, which typically includes options for cable testing, switch identification, and network verification.
2. **Testing a Network Jack:**
    - **Connect the Fluke Device:** Plug the Fluke device into the network jack you want to test using an Ethernet cable.
    - **Select the Cable Test Option:** From the main menu, select the option to test the cable. The device will analyze the connection and provide information about the cable's status.
    - **Interpret the Results:** The Fluke device will display details such as continuity, wire mapping, and any detected faults. It can identify issues like open pairs, shorts, and miswiring.
3. **Identifying VLAN Assignments:**
    - **Access the Switch Information:** After connecting the Fluke to a network jack, select the "Switch" option from the menu.
    - **View VLAN Details:** The device will retrieve and display information about the switch port and the VLAN to which the network jack is assigned.
    - **Common VLANs:** For example, on campus, you might see:
        - **VLAN 300:** Admin access
        - **VLAN 260:** Student access
        - **VLAN 220:** Faculty and staff access

    - **Troubleshoot VLAN Issues:** If a device connected to the network jack is experiencing connectivity problems, verifying the VLAN assignment can help determine if the issue is due to an incorrect VLAN.
4. **Verifying Cable Integrity:**
    - **Connect the Cable:** Attach the Ethernet cable you want to test to the Fluke device.
    - **Run a Cable Test:** Select the cable testing option, and the Fluke device will check the integrity of the cable, displaying results such as length, wire map, and any detected errors.
    - **Identify Issues:** If the cable has issues, such as breaks or improper wiring, the Fluke will identify these, allowing you to take corrective action.
5. **Documenting Results:**
    - **Record the Findings:** The Fluke device often allows you to save or record the test results. This documentation is crucial for maintaining records of network integrity and for future troubleshooting.
    - **Report Issues:** If you identify any problems during testing, document them and report them to the relevant team or supervisor for further action.

Best Practices

- **Regular Testing:** Use the Fluke device regularly to ensure network jacks and cables are functioning correctly. This proactive approach helps prevent network issues before they impact users.
- **Update Firmware:** Ensure the Fluke device's firmware is up to date to take advantage of the latest features and improvements.
- **Handle with Care:** The Fluke device is a sensitive piece of equipment. Handle it with care, store it properly, and ensure it's calibrated and maintained regularly.

Conclusion
The Fluke Network Device is a powerful tool that enhances your ability to diagnose and resolve network issues efficiently. Understanding how to use it effectively will significantly contribute to the reliability and performance of the network infrastructure you manage. Regular practice and familiarity with the device's features will make you more proficient in handling network challenges.

Installing a New Cisco Switch into a Switch Stack

Overview
Installing a new Cisco switch into a switch stack is a critical task that ensures the network infrastructure is expanded and remains stable. This chapter will guide you through the process of

installing a new Cisco switch, likely a Cisco 9260 model, into an existing switch stack. It will also explain the reasoning behind the order in which switches are stacked.

Step-by-Step Process for Installing a New Cisco Switch

1. **Physical Installation:**
    - **Power Connection:** Start by physically installing the switch into the rack. Ensure that it is securely mounted.
    - **Powering On:** Plug the switch into an electrical outlet using the appropriate power cable. Once connected, power on the switch.
    - **Stacking Cable Connection:** If the switch is being added to a stack, connect the stacking cables to the appropriate ports on the switch. These are usually located on the rear of the device.
2. **Connecting to a Computer:**
    - **Console Connection:** Use a console cable to connect the switch to your computer. The console cable typically connects to the console port on the switch and the USB or serial port on your computer.

Accessing via SSH: On your computer, open a terminal or SSH client. Use the following command to access the switch via SSH:

ssh admin@[IP address of the switch]

    -
    - **Initial Configuration:** If this is the first time the switch is being configured, you may need to set an initial IP address to access the switch via the network.
3. **Configuring the Switch:**
    - **Copying Configuration:** To maintain consistency in the network, copy the configuration from an existing switch in the stack. This is done by accessing the configuration file of another switch and pasting it into the new switch's configuration.

Use the following command to enter configuration mode:

configure terminal

To copy the configuration from another switch:

copy running-config startup-config

    - **Editing the Configuration:** Adjust the configuration settings as necessary to reflect the new switch's position in the stack, such as setting the appropriate switch number.
4. **Setting the Switch Order in the Stack:**
    - **Stack Numbering:** The switch stack is typically arranged with the oldest switch at the bottom (Switch 1) and the newest switch at the top (e.g., Switch 5 if there are five switches). This order helps maintain a clear and logical organization within the stack.
    - **Reasoning:** This practice ensures that the newest, most capable hardware is on top, potentially allowing for easier physical access for future expansions or replacements. Additionally, in some cases, the stack master (the switch that manages the stack) is often one of the more reliable or newer switches, as it may handle more intensive tasks.
5. **Finalizing the Configuration:**
    - **Applying the Configuration:**

After copying and adjusting the configuration, apply the settings by exiting the configuration mode:
end

To ensure the switch operates correctly within the stack, use the following commands to reset interfaces if needed:

interface range [interface-id]

shut

no shut

        - These commands disable and then re-enable the interfaces, ensuring that all changes take effect.
6. **Testing and Verification:**

Check the Status: After the configuration is applied, verify that the switch is functioning correctly within the stack. Use the following command to check the status of the stack:

show switch

- **Test Network Connectivity:** Ensure that devices connected through the new switch have proper network access and that the switch is communicating correctly with other devices in the stack.

Conclusion
Installing a new Cisco switch into a switch stack involves both physical and configuration steps to ensure seamless integration with the existing network. Understanding the importance of switch stacking order and following best practices ensures a stable and efficient network infrastructure. By following these steps, you can confidently add new switches to the network and maintain a well-organized switch stack.

Verifying Connections and Renaming Descriptions After Installing a New Switch

Overview
Once a new switch has been successfully installed into a switch stack, it's crucial to ensure that all other switches in the stack are connected correctly to their assigned patch panels. Additionally, updating the descriptions in the switch configurations to reflect any changes made during the installation process is essential for maintaining accurate network documentation.

Step-by-Step Process

1. **Verifying Patch Panel Connections:**
   - **Check Physical Connections:** After the new switch is installed, physically inspect all switches in the stack to ensure that they are connected to the correct ports on the patch panel. Each switch should have its cables neatly connected to the assigned patch panel ports, corresponding to the network design.
   - **Label Verification:** Verify that all cables are labeled correctly and match the documentation or network map. This helps in identifying any discrepancies or incorrect connections.
2. **Testing Connectivity:**
   - **Test Each Connection:** Use a Fluke network tester or similar device to test the connectivity of each port on the switches. Ensure that all connections provide the expected network access and that there are no loose or faulty cables.

Check VLAN Assignments: Confirm that each port is assigned to the correct VLAN by using the following SSH command:

show interface status

    - This command provides an overview of the port status, VLAN assignments, and any errors that may need to be addressed.
3. **Accessing the Switch via SSH:**

SSH Access: Log into the switch stack using SSH to access the switch configuration. Use the following command to connect:

ssh admin@[IP address of the switch stack]

Once connected, enter configuration mode to make changes:

configure terminal

4. **Renaming Descriptions:**
    - **Identify Changes:** Review any changes that occurred during the installation, such as new ports being used, cables being moved, or different VLAN assignments. These changes should be reflected in the switch's configuration.

Renaming Ports: Update the descriptions of the affected ports to match their new roles. For example, if a port's connection has changed, use the following command to rename it:

interface [interface-id]

description [New Description]

Save the Configuration: After making the necessary updates, save the configuration to ensure the changes persist:

copy running-config startup-config

5. **Documenting Changes:**
    - **Update Network Documentation:** Document any changes made during the installation process, including updates to port descriptions, patch panel connections, and VLAN assignments. This documentation is crucial for future troubleshooting and network management.
    - **Communicate Changes:** If necessary, inform the relevant team members about the changes made to the network configuration, especially if these changes impact network performance or access.
6. **Final Checks:**
    - **Run Diagnostics:** Perform a final check of the switch stack to ensure all switches are operating correctly and that there are no issues with network connectivity.
    - **Monitor Performance:** In the days following the installation, monitor the network performance to ensure that the changes have been successfully integrated and that there are no unforeseen issues.

Conclusion

After installing a new switch into a stack, verifying the connections to the patch panel and updating the switch configurations are essential steps to ensure a smooth integration. Properly renaming and documenting the changes made during the process helps maintain an organized and efficient network, reducing the likelihood of future issues.